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RESEARCH ARTICLE

A blessing in disguise?: Assessing the impact of 2010–2011 floods on trust in Pakistan*

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Abstract

The existing literature concludes that natural disasters have cataclysmic effects on the economic well-being of people. But how natural disasters shape psychological behavior is still ambiguous. In this paper, we study the effects of a natural disaster on individuals' interpersonal trust in Pakistan. By exploiting a nationally representative cross-sectional survey and combining it with the district-level geospatial data on floods, we observe the positive and statistically significant impact of 2010–2011 flooding on the reported levels of trust, after 23–36 months of the exposure. Quantitatively, a 1% increase in district's exposure to flooding increases interpersonal trust by 5%, after controlling for individual and district level characteristics. Our empirical strategy enables us to control for various geographical characteristics to counter the concerns regarding the otherwise exogenous natural experiment. Next, our results suggest that flooding heterogeneously affects individuals who are male, live in urban areas, and have higher educational attainment. These findings remain robust to different sensitivity and robustness checks.

Keywords Natural disaster · Flooding · Trust · Pakistan

JEL classifications D91 · J15 · Z13

Introduction

The existing economic research suggests that traumatic events[1] adversely affect the economic lives of people (Neumayer & Plümper, 2007; Malmendier and Nagel, 2011; Ghorpade et al., 2012; Wu et al. 2012; Calo-Blanco et al. 2017; Andrabi and Das, 2017; Abatayo and Lynham, 2019; Bucciol and Zarri, 2020). As noted by Bucciol and Zarri (2020), exposures to idiosyncratic traumatic events (such as child death, life-threatening illness, and accidents) have a negative impact on individuals' self-reported life satisfaction. Likewise, Cassar et al. (2017) and Kosec and Mo (2017) report the adverse effects of a general traumatic event (i.e., natural disaster) on an individual's economic outcomes such as aspirations, risk, and time preferences.

Nevertheless, the existing psychological evidence does not share a similar consensus when it comes to the potential impacts of traumatic events on human behavior. On the one hand, studies find that such events trigger the prosocial behavior such as social capital (Afzal et al., 2015), interpersonal trust (Cassar et al., 2017; Andrabi and Das, 2017), and political participation (Mueller et al., 2014; Fair et al., 2017), while on the other hand, studies also report the adverse effects of disasters and violence on individuals' prosocial behavior (Cameron and Shah, 2015; Albrecht, 2018).

This paper contributes to the exciting debate on the potential impacts of traumatic events by documenting the effects of natural disasters on interpersonal trust in Pakistan, the country which ranked 5th in the long-term climate risk index (CRI)

[1] The traumatic events can either be idiosyncratic (such as death of a family member or bankruptcy) or covariate (such as natural disaster or violence).

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calculated by the global climate risk index (2020). Precisely, we examine the impact of 2010–2011 flooding on interpersonal trust in Pakistan. The 2010–2011 floods are the most devastating floods in the history of Pakistan which resulted more than 2000 deaths, affected 20.4 million people, and demolished nearly 1.8 million families. This flooding was triggered by an infrequent monsoon rainstorm that released factually unprecedented levels of humidity on the mountainous northwest regions of the country.

We chose to study the effects of flooding on trust for two reasons. First, trust plays a pivotal role in the economy. At a macroeconomic level, trust positively affects economic growth (Knack and Keefer, 1997; Algan and Cahuc, 2010), welfare state support (Daniele and Geys, 2015), trade (Guiso et al., 2009), financial development (Guiso et al., 2004), and market development (Cassar et al., 2013). Nunn et al. (2018) find that the "economic downturns are more likely to cause political turnover in countries that have lower levels of generalized trust." At the microeconomic level, trust plays a significant role in explaining individuals' income differences (Butler et al. (2016)) and poverty (Collier (2002)) and also affects an individual's investment decisions (Bottazzi et al. (2016)). Second, the present literature fails to provide conclusive evidence on the impact of natural disasters on interpersonal trust.

We analyze the impact of flooding on trust in Pakistan by eliciting the data from a nationally representative survey executed by Fair et al. (2015) after 23–36 months of 2010–2011 floods. By obtaining the publicly available satellite-derived flood data in GIS vector format from United Nations Institute for Training and Research's (UNITAR) Operational Satellite Applications Program (UNOSAT)[2] and subsequently matching these district-level flood exposure data with the cross-sectional survey data, we analyze the impact of flooding on trust in Pakistan. Our results show that the district's "percentage exposure to flooding" has a positive and statistically significant impact on interpersonal trust,[3] measured by asking the question, "Presume you are walking down the street and you drop a 100 rupee note. How likely is that the following people would return the money to you?" Quantitatively, a percentage increase in districts' exposure to flooding positively affects the aggregated trust by 5%.

This finding remains robust to the inclusion of various control variables in parsimonious regression. We also use different components of our aggregated trust measure (such as trust in neighbors, family, Pakistanis, and foreigners) as the outcome variable and find that in terms of magnitude, flooding has the most prominent impact on trust in foreigners. This result seems plausible and is also consistent with the fact that foreign government and international organizations provide massive flood relief (both in terms of logistics and cash transfers) to the affected families.[4]

We also ask whether flooding affects the individuals' trust heterogeneously. To examine this, we divide our sample into subgroups based on gender, locality, and educational background. These results show that the effects of flooding on trust levels of individuals who are male, live in urban areas, and have higher educational attainment are more quantitatively sizeable compared to individuals who are female, live in rural areas, and have lower educational attainment.

As existing literature on natural disasters treats flooding as exogenous, we follow this conventional wisdom and treat 2010–2011 flooding like a natural experiment. Nonetheless, floods can also be endogenous, at least for certain individuals. For instance, people living near rivers are more susceptible to flooding, and this vulnerability might have an effect on their behavior. If so, the interpersonal trust of individuals living near rivers might be different from those who live far from the rivers.

In order to address this issue, we use various geographical control variables such as distance to the nearest river, dummies to capture whether individuals' district is located near the river, and elevation to model the possible impact of these geographical factors on human behavior. Moreover, we also control our regressions for ex-ante flood risk. Our findings remain robust to the inclusion of these geographical controls. In this manner, the empirical strategy we use in this paper also enables us to present our results more confidently as causal. Finally, we also run different sensitivity analyses such as the sensitivity of results to subgroups, sensitivity to change in outcome and treatment variables, and sensitivity to flooding induced (or, endogenous) migrations. Our baseline evidence remains robust to these sensitivity checks.

The rest of the paper is organized as follows: Section 2 provides the literature review; Section 3 is on the background of floods in Pakistan. Likewise, Section 3 explains the data and construction of variables; Section 4 introduces the empirical strategy to identify the causal impact of flooding on trust. Finally, Section 5 provides and discusses results, while section 6 concludes the study.

Natural disasters and trust: a review of literature

Trust is recognized as one of the most significant factors of modern life. Trust has several senses with more diverse

[2] United Nations Institute for Training and Research (2003), http://floods.unosat.org/ geoportal/catalog/main/home.page/

[3] Fair et al. (2017) also use the similar methodology to capture the impact of flooding on political engagements in Pakistan.

[4] Andrabi and Das (2017) also find the positive impact of 2005 earthquakes on trust in Europeans and Americans and also attribute the positive impact to relief efforts and foreign aid by Europeans and Americans.

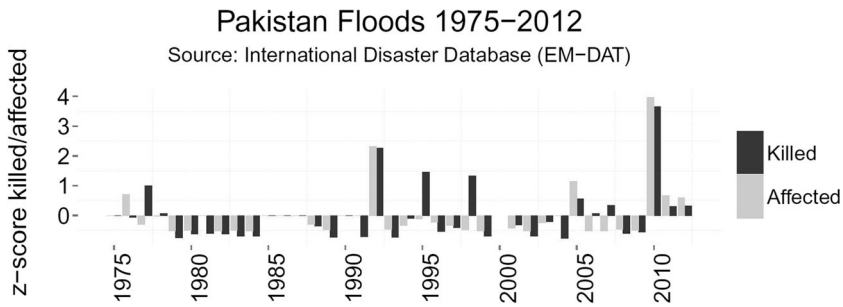

Fig. 1 Killed and Affected During 1975–2012. Source: the figure is obtained from Fair et al. (2017)

implications. Occasionally, trust point towards some social life aspects and frequently indicates some personal life characteristics. A couple of studies reflect trust as such an essential factor that proposes the existence of the humanoid groups and communities is depending upon it. The level of trust varies from one community to another, and numerous explanations have been made in this respect (Coleman, 1990). Trust is also a key component of the social capital of a community.

The concept trust responds to the conventional "homo economicus" notion by concentrating on the self-execution of norms, for instance, safety in the roads, instead of the severe maximization of profit for one's own benefit (Svendsen and Svendsen, 2003). Generally, the trust process may be considered an effort to combine economics and social psychology. Coleman (1988) first pointed out that trust is an anticipated behavior with respect to the norm that, again, empowers human's capability to collaborate on attaining predetermined shared objectives. This unpaid collaboration grounded on trust is self-enforcing and inaugurates a casual institute without any transcribed rule-books in contrast to paid support, which is imposed by a third party following any predetermined guidelines of an official institute. These conclusions can be traced back to David Hume's analysis (Olson, 1965; Ostrom, 1998; Moran and Ostrom, 2005). Fukuyama (1995) describes, "Trust proliferates when a society shares a set of ethical standards in such a manner as to establish regular expectations of honest and consistent behavior," To put it simply, when trust is developed, predictable and consistent behavior prevails among people, and the probability of being cheated significantly falls. According to Svendsen and Svendsen (2003), in the real world, the actual trust level can be labeled as the percentage of people answering yes to a very simple question "Do you think that people can be trusted, or can't you be too careful?"

Currently, there is a plethora of empirical studies available that endeavors to explain the deviation in trust levels among nations. Generally, these researches point to religious composition, ethnic variety, and income inequality as key determining factors of trust. A strand of works find that communities with hierarchical religious convictions, i.e., Islam, orthodox Christianity, and Catholicism, are less trusting (La Porta, 1997; Berggren and Jordahl, 2006). Likewise, Finseraas and Jakobsson (2012) find that nations with larger ethnic variety are occasionally found to display a lesser amount of trust. Whereas income inequality is commonly a robust determining factor of trust, carefulness must be engaged with econometric estimation and explanation as income inequality is theoretically endogenously determined. Several researchers (Moran and Ostrom, 2005; VanWey et al., 2005; Ostrom and Cox, 2010; Brondízio and Moran, 2012) have studied the human-environment interaction and behavioral theory linkages with trust from a variety of perspectives (Ostrom & Walker, 2003; Ostrom, 2003; Ostrom, 2009; Brondízio and Moran, 2012). This literature has reflected the role that the natural environment may play in shaping trust. A unique question originated from this discussion: Is the public influenced by the occurrence of natural tragedies, and if so, how?

A strand of the literature suggests that traumatic events such as natural disasters damage or destroy human and physical capital in a short period but researchers have also found that they have progressive spillover impacts. Firstly, Toya and Skidmore (2014) pointed out that the frequency of environmental tragedies is positively associated with human capital growth, TFP, and economic development in a long

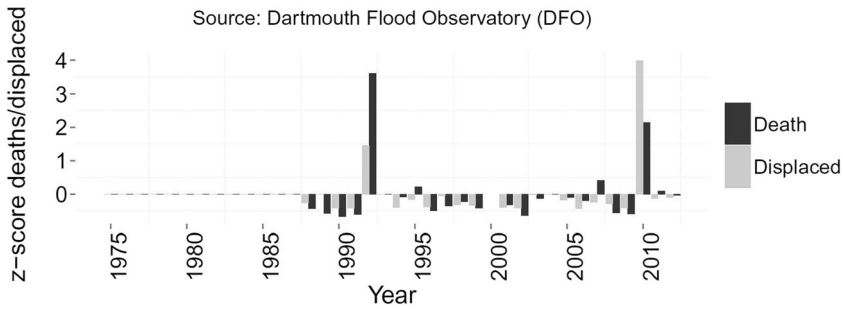

Fig. 2 Death and displaced during 1975–2012. Source: the figure is obtained from Fair et al. (2017)

Table 1 Descriptive statistics of outcome variables

Variable(s)	Mean	Std. Dev.	Min	Max
Trust (aggregate)	10.54	2.54	4	16
Neighbor	2.92	1.06	1	4
Family member	3.69	0.68	1	4
Pakistani	2.07	0.93	1	4
Foreigner	1.85	0.95	1	4

The descriptive statistics of the measure of trust quantified using the question, "Presume you are walking down the street and you drop a 100 rupee note. How likely is that the following people would return the money to you?"

Data source: Fair et al. (2015)

Table 2 Descriptive statistics of explanatory variables

Variable(s)	Mean	Std. Dev.	Min	Max
% Pop. exposed	0.090	0.137	0.000	0.692
% Area exposed	0.126	0.171	0.000	0.753

Data source: United Nations Institute for Training and Research (2003) and Oak Ridge National Laboratory (2008)

period of time. They claimed that natural tragedies usually damage physical capital in a short period but in the rescue procedure, the creative destruction idea of Schumpeter can take place, in which tragedies offer a chance to modernize the capital standard and encourage the implementation of new technologies. Important to mention, the researchers revealed that ecological calamities are positively linked with economic development but the opposite is observed in geologic calamities. While the expansion of experimental methods has supported some estimates, geographic threats are still rather ambiguous (Crespo Cuaresma et al., 2008; Leiter et al., 2009). These proposals are true for some cases but cannot be generalized as a rule of thumb. For instance, the Great Earthquake Japan (2011) impacts are not faded after a decade, while the Japanese government highlighted the threat of another tsunami further crippling the already-wrecked Fukushima Daiichi nuclear power station.

In recent decades, the scale of destruction following natural disasters has significantly increased, a bulk of studies have emphasized the consequences of these tragedies in economic development and behavioral context (Belter & Shannon, 1993; Ursano et al., 2008; Han et al., 2017). Researchers are desperate to analyze the impact of the natural disasters on a number of micro and macro perspectives, for instance, trust, risk aversion, human capital, migration, and national income (Yang, 2008; Noy, 2009; Cameron and Shah, 2010; Baez et al., 2010; Fleming et al., 2014). Apart from structural and economic losses, societies affected by traumatic events like floods (Wu et al., 2008; Dottori et al., 2018) also undergo a loss in trust, which can directly influence their social cohesion, trust level, and perceptions about democratic norms both in short and long terms. In addition, trust loss leads to disturb and slow down the recovery of flood-afflicted societies (Liang, 2016; Han et al., 2011; Han et al., 2020).

To examine the proposition that natural disasters improve trust, Toya and Skidmore (2014) conducted a panel data analysis for more than 140 countries. A positive association between levels of trust and natural disaster frequency is found. It is attention-grabbing that this association was obvious purely for storms (among volcanic eruptions, earthquakes, floods, storms, etc.). Authors put forward that unlike other natural tragedies, storms disturb everybody in the region irrespective of wage level and societies' effort collectively to address their challenges, as a result, establish trust.

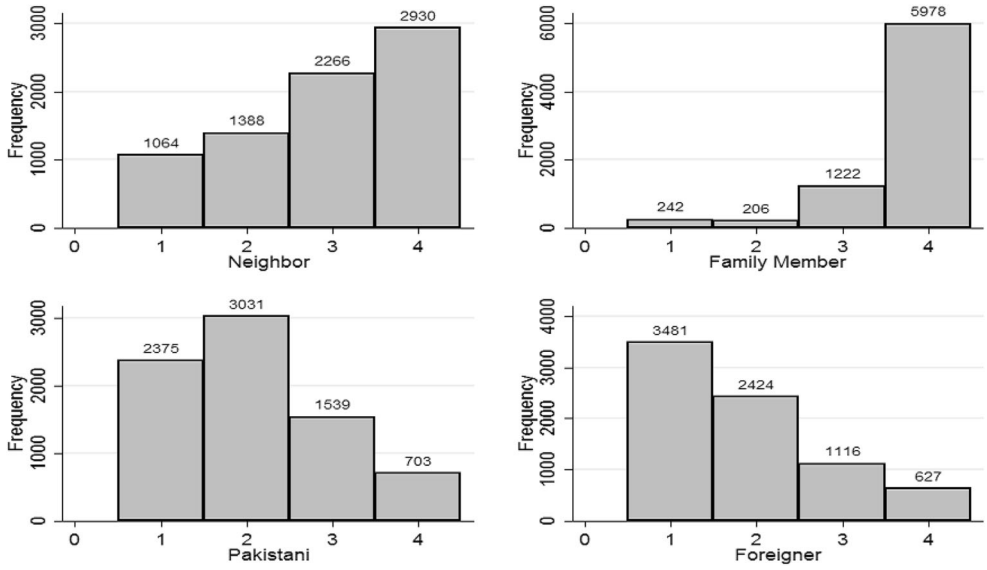

Fig. 3 Measure of trust. Graph is based on the question: "Presume you are walking down the street and you drop a 100 rupee note. How likely is that the following people would return the money to you?". The answer is coded as "1. Would not happen; 2. Not very likely; 3. Somewhat likely; 4. Very likely." Each figure provides the frequency of the population answering to these options. Data source: Fair et al., (2015)

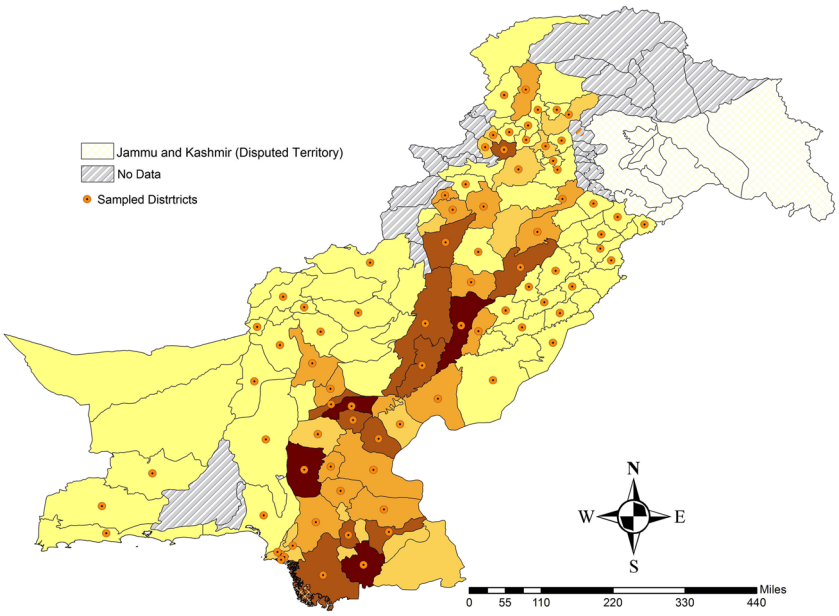

Fig. 4 Percentage of population affected by 2010–11 Floods. Author's own calculations using data from the United Nations Institute for Training and Research (2003) and Oak Ridge National Laboratory (2008). Each polygon represents the district, the second-largest administrative unit within Pakistan after provinces. The darker the polygon, the more the population exposed to 2010–2011 floods (in percent). The orange-colored dots in polygons represent sampled districts (Fig. 5)

Toya and Skidmore's (2014) findings are further supported by some recent studies. Cassar et al. (2017) found evidence that the sufferers of the Indian Ocean tsunami (2004) turned out to be more impatient, risk-averse, and trusting in a long period of time. Additionally, Shupp et al. (2017) showed that for individuals who were victimized by a tornado, trust in general, trust in friends, and trust in the police and fire authorities augmented. They further emphasized that victimized property holders became less tolerant on the subject of financial payments than un-victimized property holders. Calo-Blanco et al. (2017) showed that social cohesion is positively affected by exposure to the earthquake in Chile. They also revealed that social cohesion greater than before after a high scale earthquake but gradually faded as ecological situations developed with the passage of time. Dussaillant and Guzm'an (2014) study also found a positive influence of the earthquake on trust levels in Chile after the 2010 earthquake. The authors also pointed out that the association between trust level and earthquake was significantly dependent on the past ranks of societal trust. On the other hand, a strand of literature (Mueller et al., 2014; Papanikolaou et al., 2012; Ahsan, 2014; Fleming et al., 2014) also show empirical evidence that natural tragedies have no impacts or negative impacts on the levels of trust in a community. Even though numerous studies showed that natural tragedies affect core preferences, the route of the effect differs. In order to better recognize the association between natural disasters and trust, this paper sheds light on the impact of 2010–2011 flooding on interpersonal trust using a unique cross-sectional survey from Pakistan.

A brief history of 2010–2011 floods in Pakistan

Floods can be recognized as the major natural disaster in the history of Pakistan. Since its inception, Pakistan has witnessed around 72 floods of varying intensities in the last five decades. In July 2010, following abnormally heavy monsoon rains, Pakistan experienced yet another worst floods in history. Figures 1 and 2 provide a visual display of standardized values for the number of killed, displaced, and affected for each flood covering the period of 1975–2015. Figure 1 presents data from Dartmouth Flood Observatory (DFO), and the second graph draws on the data from the International Disaster Database (EM-DAT). Indicators report that more than 1/10 of the total population was affected and/or temporarily displaced by the 2010 flooding which is regarded as the largest in the entire history of Pakistan. In terms of infrastructure, 78 out of 141 districts were severely destroyed by this natural calamity. The Ex-UN Secretary-General Ban Ki-moon pointed out the demolition triggered by the 2010 flood by saying that, "in the past, I have witnessed many natural disasters around the world, but nothing like this."[5]

Mega tragedy needs a mega response, but compared with responses to other crises, the instantaneous global funding response was slow this time. As per the World Bank report (2010:24), costs associated with flood destruction are projected at nearly $8.8 to $11 billion including short-term and long-term rebuilding and costs for relief. In order to cope up with this catastrophe, the Pakistan government appeals to the international community for financial and physical support. The UN arranged immediate assistance like pure water,

[5] See, https://www.theguardian.com/world/2010/aug/15/pakistan-floods-united-nations-aid/.

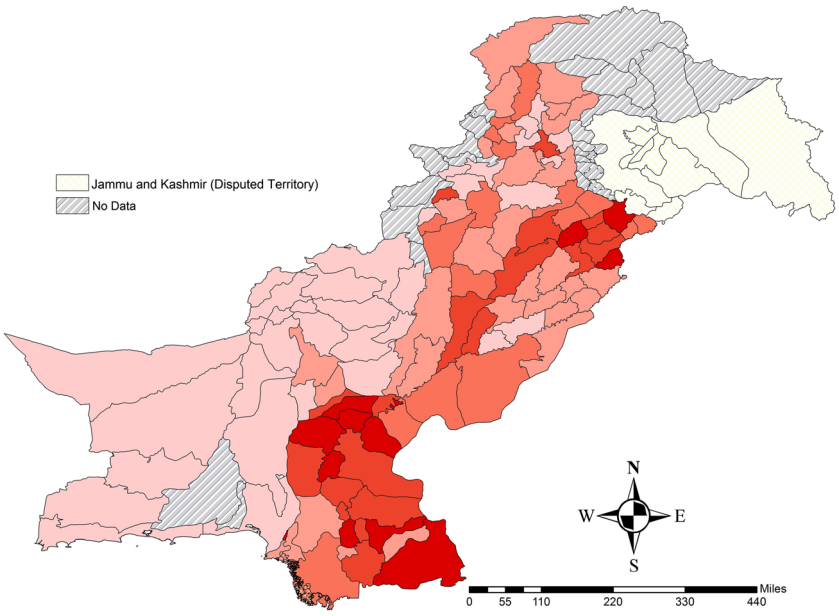

Fig. 5 Ex-ante flood risk (0–5). We aggregate tehsil (sub-district) level data of ex-ante flood risk index provided by Fair et al. (2017) at the district level. Each polygon represents the district, the second-largest administrative unit within Pakistan after provinces. Darker the polygon, the higher the flood risk

clothes, shelter, and foodstuff via launching a relief package worth $460 million.

Foreign government and NGO's provided equipment, sent doctors, and donated supplies and money to complement the relief efforts of Pakistani governments and civilians. According to the United Nations (2010), almost $1.8 billion had been devoted to flood charitable funding including 30.7% by the USA, 17.5% by individuals and other private organizations, and 13.5% by Saudi Arabia.

Due to the strategic importance of Pakistan, the US government was keenly interested to help the affected areas through flood relief packages. To placate the distorted views of Pakistani people regarding US governments stemming from counterinsurgency practices together with drone attacks in the country, the USA generously helped Pakistan by providing the largest and very first supplies of food and shelter-related items. In addition to the nearly 1 billion dollars donated through other networks since mid-2010, some $500 million of Kerry-Lugar bill was also approved for faster distribution for flood assistance by cash transfers to the affected families. The US government had wished to improve its image among the Pakistani public, which is why direct payment cards were branded with the US flag (Fair et al., 2017).

From the perspective of immediate flood relief, self-supporting activities emerged at the initial stages of tragedy and continued all the way through. Controversial Islamic organizations like Jamat-E-Islami and Jamat-ut-Dawa had been active through its AlKhidmat Foundation and played a prominent role in the recovery process. These organizations had certainly gained a unique experience from their involvement in the Kashmir earthquake response (2005), which is why their relief activities were witnessed to have a significant reach in the early weeks of the flood. Moreover, motivated by localized self-help efforts, civil society divided the volunteers into different groups at very local levels like villages, administrative councils, and *mohallas* (streets). The prime aim of these groups was to collect and distribute donations from local and international organizations. Accompanied by these efforts, even the chamber of commerce, professional bodies, and individual philanthropists donated supplies and money to the affected people. These local forms of volunteerism were not unique to the 2010 flood, but similar high-level efforts are documented following the 2005 earthquake of northern regions.

Data and construction of variables

Outcome variable: individual-level trust

We derive the data on trust from a national representative survey fielded by Fair et al. (2015) in 2013. The survey is based on 7656 randomly selected adult individuals from several districts of Pakistan. This survey is conducted by Case Western Reserve University under the project titled, "Pakistani Media and Muslim Communities: Identifying and Supporting Critical Thinking," using the expertise of a Pakistani non-government organization named Socio-Economic Development Consultants (SEDCO). Primarily, this survey gathers the information related to Pakistani people's attitudes towards drone attacks in tribal areas. However, out of 108 survey items, the question related to trust is also asked from the respondents.

The question we exploit to quantify trust asks, "Presume you are walking down the street and you drop a 100 rupee note. How likely is that the following people would return the

Table 3 Descriptive statistics of control variables

Variable(s)	Mean	Std. Dev.	Min	Max
Geographical controls				
Flood risk (0–5)	1.163	0.929	0	4.339
Distance to maj. river	0.873	1.207	0.058	5.511
Near maj. rivers	0.595	0.494	0	1
Neighboring maj. river	0.643	0.482	0	1
Elevation (mean)	0.160	0.206	0.002	0.765
Elevation (std.)	0.039	0.054	0	0.249
% Pop. exposed (flood 2012)	0.067	0.226	0	1.551
Crop-suitability index (0–5)	1.667	0.963	0	5
Individual-level demographic and economic controls				
Urban	0.41	0.49	0	1
Female	0.50	0.50	0	1
Married	0.74	0.44	0	1
Age (years)	33.76	12.13	18	90
Age squared	1286.58	956.90	324	8100
Education (all categories)	1.73	1.65	0	6
Illiterate	0.36	0.48	0	1
Primary	0.13	0.34	0	1
Middle	0.15	0.36	0	1
Matric	0.20	0.40	0	1
Intermediate	0.09	0.29	0	1
Bachelors	0.05	0.22	0	1
Professionals	0.01	0.11	0	1
Wealth index	0.75	0.38	0	3.91
Residence status	0.89	0.31	0	1
Nominal income (1000 Rs.)	2.11	3.84	0	250
Earn more	0.13	0.34	0	1
Earn less	0.44	0.50	0	1

Distance to maj. river, near maj. rivers, and neighboring maj. river capture the location of given district relative to the river. Distance to maj. river is the continuous variable which measures the distance from the center of given district to major river. However, near maj. rivers, and neighboring maj. river are dummy variables where the former is equal to one if river is within the 50-km radius of district, while the latter variable is equal to unity when district boundary is adjacent to the major river

Data source: United Nations Institute for Training and Research (2003), Oak Ridge National Laboratory (2008), Fair et al. (2015), and Fair et al. (2017)

money to you?" Having four distinct categories ranging from very likely to will not happen. Each question measures trust in family, neighbor, Pakistani, and foreigner.[6] We make an additive index of trust to assess the impact of flooding on trust.

Following the sampling frame developed and adopted by the Pakistan Bureau of Statistics (PBS) to conduct a national survey, Fair et al. (2015) randomly select each respondent from 479 primary sampling units (PSUs). As per the shares of rural and urban areas in total population according to the 1998 census, 194 of these PSUs are rural and 285 are urban. The district representative sample was collected from Pakistan between the months of August and October in 2013. Fair et al. (2015) reported, "The overall response rate for the survey was 64% (67% in Balochistan, 63% in KP, 65% in Punjab, 61% in Sindh province), which is lower than the 70% obtained in the General Social Survey in recent years but higher than the 59.5% achieved by the American National Election Studies (2008)[7] and other high-quality academic studies" (Fair et al., 2015, p. 857). Table 1 gives the descriptive statistics of our measure of trust. On average, respondents report the highest trust in Family Members and the lowest trust in foreigners. Figure 3 shows the histogram of components of our trust measure.

Treatment variable: district-level flood exposure in 2010–2011

We rely on geospatial data on floods to construct our treatment variable at the district level. We obtained publicly available satellite-derived flood data in GIS vector format from the United Nations Institute for Training and Research's (UNITAR) Operational Satellite Applications Program (UNOSAT).[8] Following Fair et al. (2017), we then acquired ORNL's LandScan population data available at 1 km (30″ × 30″) spatial resolution[9] and used this information to calculate the percentage of population exposed to 2010–2011 floods at the district-level in Pakistan.[10] Table 2 offers the descriptive statistics of our treatment variables. Similarly, Fig. 4 depicts the percentage of the population exposed to 2010–2011 floods.

Control variables

Controlling the regressions for relevant covariates is indispensable to obviate the threat of omitted variable bias. We add a set of control variables in our regressions. We start by adding ex-ante flood risk obtained from Fair et al. (2017). We also control our regressions for the district's proximity to a major river. Geographic controls such as mean and standard

[6] World Values Survey (2016) uses the similar methodology to measure trust in their surveys. Similarly, Andrabi and Das (2017) also use the same question to quantify the trust while assessing the impact of 2005 earthquake on trust in Northern Pakistan.

[7] American National Election Studies (2008), www.electionstudies.org/studypages/2008prepost/2008prepost.html

[8] United Nations Institute for Training and Research (2003), http://floods.unosat.org/geoportal/catalog/main/home.page

[9] Oak Ridge National Laboratory (2008), https://landscan.ornl.gov/

[10] One caveat of using geospatial data, noted by Fair et al. (2017) is, "The UNOSAT data underestimate the floods' impacts in steep areas where the flood waters did not spread out enough to be identified with overhead imagery but where contemporaneous accounts clearly show there was major damage at the bottom of river valleys. In Upper Dir district in KPK, for example, the UNOSAT data show no flooding."

Table 4 Baseline results

	(1)	(2)	(3)	(4)	(5)
	Log of aggregate trust level				
% Pop. exposed	0.0643*** (0.020)	0.0650*** (0.020)	0.0785*** (0.018)	0.0516*** (0.019)	0.0497*** (0.016)
Observations	7648	7648	7648	7504	6953
R-squared	0.024	0.045	0.079	0.119	0.142
Province fixed effects		Y	Y	Y	Y
Geographical controls			Y	Y	Y
Demographic controls				Y	Y
Economic controls					Y

The unit of analysis is an individual. Columns 2–5 progressively expand the model by adding province fixed effects, geographic controls, and demographic and economic controls, respectively. Geographic controls include ex-ante flood risk, distance to major rivers from district centroid, a dummy for district border with a major river, elevation (mean), ruggedness, and crop-suitability index. Economic controls include wealth index, residence status, nominal income, and indicator variable each for earning more or earning less than the median income

In all the regressions, standard errors (given in square brackets) are clustered at district level, where *** < 0.01; ** < 0.05; and * < 0.10

deviation of elevation are also included as potential covariates. Finally, we include individual-level demographic and socio-economic variables as controls. The descriptive statistics of all the control variables we include in the analysis are provided in Table 3.

Empirical strategy

This section explains the empirical strategy to identify the causal impact of flooding on interpersonal trust in Pakistan. In order to examine the impact of flooding on trust, we exploit the individual-level trust data obtained from a nationally represented survey conducted by Fair et al. (2015) in fall 2013, after 23–36 months of 2010–2011 floods. We then combine these individual-level data with the district-level satellite data on flood exposure. This matching strategy is extensively used by the researchers and implicitly assumes that the floods evenly affected the individuals within the district.

In order to tear out the causal impact of flooding on interpersonal trust, our empirical analysis uses the following baseline regression equation:

$$\log(trust)_i = \alpha + \beta\,(flooding)_d + \gamma_p + BX_i + CZ_{\mathrm{d}} + \varepsilon_i \quad (1)$$

where $\log(trust)_i$ is a log of the additive trust index for *ith* individual, $(flooding)_d$ represents the percentage population exposed to 2010–2011 floods in *dth* district, and γ_p captures the province fixed effects. Moreover, X_i is the vector of individual-level demographic and economic controls, and Z_d is the vector of district-level controls. Finally, ε_i is the error term. Equation 1 shows that β, the elasticity of trust level to flood exposure, is the coefficient of interest. As exposure to a natural disaster is exogenous, still, floods can be predicted given the proximity to major rivers. So it is expected that flood predictability at the individual-level might induce the change in behavioral parameters (such as cooperation, reciprocity) of individuals living near the major rivers. This observation has

Table 5 Disaggregated results

	(1)	(2)	(3)	(4)
	Neighbor	Family	Pakistani	Foreigner
% Pop. exposed	0.0945*** (0.022)	0.0031 (0.013)	0.0499** (0.024)	0.0822** (0.030)
Observations	6953	6953	6953	6953
R-squared	0.158	0.048	0.117	0.136
Province fixed effects	Y	Y	Y	Y
Controls	Y	Y	Y	Y

The unit of analysis is an individual. Each column includes province fixed effects, geographic controls, and demographic and economic controls. Geographic controls include ex-ante flood risk, distance to major rivers from district centroid, a dummy for district border with a major river, elevation (mean), ruggedness, and crop-suitability index. Economic controls include wealth index, residence status, nominal income, and indicator variable each for earning more or earning less than the median income. In all the regressions, standard errors (given in square brackets) are clustered at district level, where *** < 0.01; ** < 0.05; and * < 0.10

Table 6 Heterogeneous results

	Male	Female	Urban	Rural	Low edu.	Medium edu.	High edu.
% Pop. exposed	0.0637*** (0.020)	0.0298* (0.017)	0.0887*** (0.020)	0.0341 (0.021)	0.0346* (0.018)	0.0616*** (0.018)	0.0852*** (0.025)
Observations	3573	3380	2772	4181	3353	2501	1099
R-squared	0.170	0.137	0.158	0.135	0.134	0.162	0.190
Province fixed effects	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y

The unit of analysis is an individual. Each column includes province fixed effects, geographic controls, and demographic and economic controls. Geographic controls include ex-ante flood risk, distance to major rivers from district centroid, the dummy for district border with a major river, elevation (mean), ruggedness, and crop-suitability index. Economic controls include wealth index, residence status, nominal income, and indicator variable each for earning more or earning less than the median income. In all the regressions, standard errors (given in parentheses) are clustered at district level where *** < 0.01; ** < 0.05; and * < 0.10

serious implications exogeneity of floods and also for our causal interpretation of the flood impact on trust.

In order to overcome the given predictable nature of floods, we control our regressions for geographical variables such as the district's proximity to major rivers, dummy variables indicating whether the district neighbors river, elevation, and ex-ante flood risk index. Controlling for the geographical variables counter the concerns regarding the endogeneity of flooding and also enable us to present our results more confidently as causal.

Another immediate concern regarding our empirical strategy, which relies on cross-sectional data, is that our estimated results might be under the influence of endogenous migration. One can argue that natural disasters induce massive displacements and non-migrants are not the representative of the ex-ante population. If it is the case, our estimated results could either overstate or understate the true impact of flooding on trust. Interestingly, the timing of fielded surveys with respect to 2010–2011 flooding resolves the problem of endogenous migration. As Mueller et al. (2014) report, "modest to insignificant impacts" of flooding on long-term migration in the case of Pakistan. If so, we expect the flood-induced displaced population to return their homes in the span of 21–36 months, when the survey was fielded. Besides, we also control for the district-level internal migration estimates as a robustness check.

Table 7 Subgroups on the basis of the district's proximity to rivers

	(1) Near maj. rivers	(2) Neighboring maj. river
% Pop. exposed	0.0485*** (0.017)	0.0591*** (0.019)
Observations	5892	4880
R-squared	0.149	0.167
Province fixed effects	Y	Y
Controls	Y	Y

Unit of analysis is an individual. Each column includes province fixed effects, geographic controls, and demographic and economic controls. In all the regressions, standard errors (given in parentheses) are clustered at district level where *** < 0.01; ** < 0.05; and * < 0.10

Results and discussion

Baseline results

First, we investigate the impact of flood exposure on trust levels by estimating Eq. 1 using OLS. Table 4 shows the effects of flooding (measured by, percentage of population exposed to flood in a given district) on the trust levels of the individuals. Column (1) provides the raw correlation between flood exposure and trust levels. Columns (2)–(4) progressively expand the baseline model by incorporating province fixed effects, geographical controls, and individual-level socioeconomic and demographic controls, respectively. Interestingly, the baseline results show that flood exposure is positively associated with trust levels. Based on the fully specified econometric specification (column 6 of Table 4), individuals living in the district with 1% more flood exposure on average report 5% higher trust. This result remains significant for all the specifications.

Our findings are consistent with those of Andrabi and Das (2017) which shows the positive effects of exposure to the earthquake on trust in Europeans and Americans. Similarly, Fair et al. (2017) and Rudolph and Kuhn (2018) also find positive effects of flooding on political engagements in Pakistan and Germany, respectively. Contrary to our findings, Afzal et al. (2015) conclude that social capital, measured by contribution in public goods, declines as a result of a severe flood, but mild experiences enhance it in selected districts of Pakistan.

Table 8 Sensitivity results: % area exposed as an alternative measure

	(1)	(2)	(3)	(4)	(5)
Log of aggregate trust level					
% Area exposed	0.0549*** (0.016)	0.0592*** (0.016)	0.0693*** (0.014)	0.0499*** (0.015)	0.0441*** (0.012)
Observations	7648	7648	7648	7504	6953
R-squared	0.027	0.051	0.087	0.121	0.144
Province fixed effects		Y	Y	Y	Y
Geographical controls			Y	Y	Y
Demographic controls				Y	Y
Economic controls					Y

The unit of analysis is an individual. Each column includes province fixed effects, geographic controls, and demographic and economic controls. Geographic controls include ex-ante flood risk, distance to major rivers from district centroid, the dummy for district border with a major river, elevation (mean), ruggedness, and crop-suitability index. Economic controls include wealth index, residence status, nominal income, and indicator variable each for earning more or earning less than the median income. In all the regressions, standard errors (given in parentheses) are clustered at district level where *** < 0.01; ** < 0.05; and * < 0.10

Disaggregated trust

The baseline evidence is based on the log of the additive trust index. Besides aggregate trust, it is also of primary interest to know how the trust in different sections of the society is affected by the exposure to flood. Table 5 shows the disaggregated results for different communities. Column (1) reports that 1% higher exposure to flood enhances trust in neighbors by 13%. The flood exposure has a negligible and statistically insignificant impact on trust in the family. Finally, trust in Pakistanis and the trust of foreigners are increased by 9% and 12%, respectively. It is worth mentioning that the analysis presented in Table 5 includes the vector of covariates which we use in the final regression of baseline results presented above in Table 4.

Heterogeneous results

This section reports the heterogeneous results across subgroups based on gender, locality, and education. Interestingly, we find that flood exposure unevenly affected individuals across the subgroups. Column (1) and (2) of Table 6 splits the results across genders.

Results indicate that in males, the magnitude of trust raised as a result of exposure to flooding is higher as compared to females. This finding is consistent with Neumayer and Pl¨umper (2007) which conclude that females are severely affected by natural disasters as compared to males. Secondly, urban inhabitants experienced a higher surge in trust levels compared to rural inhabitants. As per the education, individuals with higher education attainment report a larger increase in trust levels compared to individuals with medium or low educational attainment. The higher impact on urban and educated individuals can partly be explained by their higher media consumption: People who are exposed to media (electronic and print) will be better informed about the relief efforts and foreign aid, which inculcate prosocial behavior.

Table 9 PCA of trust as an outcome variable

	(1)	(2)	(3)	(4)	(5)
Trust level (PCA)					
% Pop. exposed	0.3670*** (0.111)	0.4015*** (0.106)	0.4342*** (0.101)	0.2998*** (0.107)	0.2795*** (0.096)
Observations	7648	7648	7648	7504	6953
R-squared	0.030	0.053	0.079	0.125	0.151
Province fixed effects		Y	Y	Y	Y
Geographical controls			Y	Y	Y
Demographic controls				Y	Y
Economic controls					Y

The unit of analysis is an individual. Each column includes province fixed effects, geographic controls, and demographic and economic controls. Geographic controls include ex-ante flood risk, distance to major rivers from district centroid, the dummy for district border with a major river, elevation (mean), ruggedness, and crop-suitability index. Economic controls include wealth index, residence status, nominal income, and indicator variable each for earning more or earning less than the median income. In all the regressions, standard errors (given in parentheses) are clustered at district level where *** < 0:01; ** < 0:05; and * < 0:10

Table 10 Controlling for migration

	(1)	(2)
% Pop. exposed	0.0497*** (0.015)	
% Area exposed		0.0485*** (0.013)
Migration (estimated)	0.0262 (0.076)	0.0340 (0.077)
Observations	6953	6953
R-squared	0.141	0.146
Province fixed effects	Y	Y
Controls	Y	Y

Unit of analysis is an individual. Each column includes province fixed effects, geographic controls, and demographic and economic controls. In all the regressions, standard errors (given in parentheses) are clustered at district level where *** < 0.01; ** < 0.05; and * < 0.10

Sensitivity analysis

This section helps strengthen the findings by assessing the sensitivity of baseline results to various changes in the analysis. First, we replicate the baseline results for subgroups on the basis of the district's proximity to major rivers to see whether the results remain comparable for individuals living in districts near major rivers. Then, we change the treatment variable from % pop. exposed to % area exposed and outcome variable to PCA of trust from aggregated trust in subsequent regressions to assess the robustness of baseline results. In the final set of regressions, we control our analyses for estimated migration to suppress the concerns of flood-induced (or, endogenous) migrations.

Sensitivity to subgroups

We provide the results of subgroups on the basis of districts' proximity to major rivers in Table 7. The results for subgroup analyses sustains our sensitivity test in a sense that contrary to our expectations, the magnitude of effect on trust is also comparable to that of our baseline results presented in Table 4. Quantitatively, a 1% increase in exposure to flooding enhances individuals' trust by 5% in the subgroup of districts near major rivers.

Sensitivity to alternative measures as outcome and treatment variables

We also assess the impact of flooding on trust levels by utilizing a percentage of areas exposed to floods in a given district as a treatment variable. Results provided in Table 8 substantiate the findings given in the above sections. Surprisingly, not only the sign but magnitude is also comparable with our preferred specification of Table 4. Quantitatively, a 1% increase in the district's area exposed to flood increased the trust levels of individuals residing in the district by 4.45%. Moreover, this finding is significant for all the specifications presented in Table 8.

We also check the sensitivity of results to the change in the form of the outcome variable. Table 9 sheds light on the estimated results in which we change the simple additive index of trust to trust measure obtained through principal component analysis (PCA). The findings from this specification are also consistent with the baseline evidence.

Controlling for migration

Finally, we include the district level estimated measures of internal migration as a robustness check. As discussed above, an immediate concern about the analysis could be that our findings are under the influence of endogenous migration. First, we hypothesize that the cost of long-run migration due to disaster is very high and most of the people return to their residents soon after the disaster ends, primarily in the case of rural areas. The timing of the survey also helps us to obviate the concerns of endogenous migration. Nonetheless, we control our regressions for estimated migrations at the district level obtained from Fair et al. (2017). Table 10 shows that controlling for migration does not affect either the sign or the magnitude of our results.

Conclusion

In this paper, we study the effects of 2010–2011 flooding on the trust level of individuals in Pakistan. By matching the nationally representative individual-level cross-sectional survey data with district-level geospatial data on 2010–2011 floods, we observe the positive and statistically significant increase in the individuals' self-reported interpersonal trust, after 23–36 months of exposure to the most destructive floods in Pakistan's history. Quantitatively, our preferred specification indicates that a 1 % increase in exposure to flooding enhances individuals' interpersonal trust by 5%. We further show that flooding has heterogeneous positive effects on individuals who are male, live in urban areas, and are graduated from college. One more interesting result of the study is that as a result of exposure to 2010–2011 flooding, trust in foreigners was enhanced by a greater amount compared to trust in neighbors, family, and Pakistanis. In particular, our finding corroborates with the existing scholarship which assesses the effects of natural disasters on prosocial behavior in Pakistan (Fair et al., 2017; Andrabi and Das, 2017). Conversely, our findings differ from that of Kosec and Mo (2017) and Afzal et al. (2015) which found that 2010–2011 flooding had negatively impacted the individuals' aspirations and social capital in Pakistan, respectively.

Broadly, this study makes two scholarly contributions to the existing literature on social psychology and behavioral

economics. First, unlike most of the existing studies, we provide evidence based on the nationally representative survey that natural disasters have a positive impact on individuals' trust in Pakistan, one of the most vulnerable countries to climate change (Global Climate Risk Index, 2018). Second, given the importance of trust in economic transactions, the positive impact of natural disaster on trust can be regarded as good news for Pakistan's domestic economy as policymakers just have to focus on physical damages done by disasters, and the effects of higher interpersonal trust on the economy might counterbalance the devastating physical damaged caused by natural disasters.

Authors' contributions Noman Ahmad: conceptualization, methodology, and software. Muhammad Zeeshan Younas: theoretical framework, analysis, and writing—reviewing and editing.

Data availability The data and material that support the findings of this study are available from the corresponding author upon reasonable request.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval and consent to participate Ethics approval was not required for this study.

Consent for publish Not applicable.

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